$$WTP_i^* = \exp(\mathbf{x}_i{}'\beta + \varepsilon_i), \quad [20]$$

and the probability of individual i responding "yes" to a specified bid amount "bid$_i$" becomes

$$\Pr(WTP_i^* \geq bid_i \mid \mathbf{x}_i{}') = 1 - F(bid_i \mid \mathbf{x}_i{}') = 1 - \Phi\left(\frac{ln(bid_i) - \mu}{\sigma}\right), \quad [21]$$

where WTP^* is the log of WTP, Φ is the standard normal cumulative distribution function, μ and σ are, respectively, the mean and standard deviation of the log transformation of WTP. Assuming $WTP_i = 1$ if the respondent is willing to pay the specified bid amount and 0 otherwise, the log likelihood function can be written as

$$ln\ L = \sum_{i=1}^{n} \left\{ (WTP_i) ln\left[1 - \Phi\left(\frac{ln(bid_i) - \mu}{\sigma}\right)\right] + (1 - WTP_i)\, ln\ \Phi\left(\frac{ln(bid_i) - \mu}{\sigma}\right)\right\}. \quad [22]$$

A probit regression model is estimated to model the determinants of the predicted probability that the individual is willing to pay the specified bid amount. An additional variable controlling for the number of individuals in the household who experienced symptoms is added to this model. Given that the respondent was valuing a 50% reduction in all symptom days experienced in the household, the duration of the illness is captured by a variable representing half of all symptom days experienced in the household. A test down approach is implemented to arrive at the final reduced model, eliminating variables one at a time that do not have a statistically significant effect on the predicted probability that the individual is willing to pay the specified bid amount. The results of this reduced regression model can be found in Table 7.[5,6]

[5] For comparison purposes, a logit, normal, and log-logit model were also estimated. The Akaike and Bayesian information criterion confirm that the log-normal and log-logit models are superior to the linear in bid models, and the log-logit and log-normal models are virtually identical in fit.

[6] The results of the full model are available from the authors.

The bid coefficient in this model is negative and statistically significant at the 1% level, indicating that the higher the bid amount, the less likely the individual is willing to pay, all else constant. This provides evidence of theoretical construct validity to the CVM question responses. The natural log of half of all household symptom days is positive and statistically significant at the 5% level. The coefficient on this variable is less than one, implying that WTP increases with household symptom days, but at a decreasing rate. This is similar to findings by Alberini et al. (1997), Johnson, Fries, and Banzhaf (1997), Liu et al. (2000), and Dickie and Messman (2004). Individuals with a current heart disease are also more likely to be willing to pay the specified bid amount, and this variable is significant at the 10% level.

Turning to lifestyle and demographic factors, we find that being a college or technical school graduate has a positive and significant effect on WTP compared to those respondents without degrees. Having health insurance has a negative effect on the probability that the individual is willing to pay the specified bid amount, and this variable is significant at the 1% level. Finally, living in the city of Duarte or Burbank has a positive and significant effect on WTP compared to living in Monrovia, Sierra Madre, or Glendora. Interestingly, a variable controlling for the number of individuals in the household who experienced symptoms was included in the full model but did not have a significant effect on WTP.

VII. BENEFIT ESTIMATES FOR A REDUCTION IN ONE WILDFIRE SMOKE–INDUCED SYMPTOM DAY

Defensive Behavior Method WTP

In the DBM regression model, the individual WTP for a given change in illness dS can be calculated as $-[p_d/(\partial S/\partial D)]dS$ from equation [13]. Given that using a home air cleaner is the only defensive action found to have a statistically significant effect on symptom days, the WTP measure is based on this action. The incremental effect of this endogenous input on output is –0.31, meaning the use of an air cleaner is expected to reduce symp-

TABLE 7
Probit Regression of WTP for 50% Reduction in Symptom Days

Variable	Coef.	Std. Err.
ln(Bid amount)	− 0.455***	0.095
ln(Half of household symptom days)	0.338**	0.156
Current heart condition	0.695*	0.365
College/technical school graduate	0.666**	0.286
Has health insurance	− 1.093***	0.391
Lives in Duarte	0.687**	0.343
Lives in Burbank	0.546*	0.294
Constant	1.201**	0.575
N	157	
Log likelihood	− 78.873	
Likelihood ratio χ^2 (7)	45.610	
Prob $> \chi^2$	0.0000001	

* $p \le 0.10$; ** $p \le 0.05$; *** $p \le 0.01$.

tom days by 0.31.[7] Taking the average cost reported by those respondents who used an air cleaner in their home during the Station Fire results in an estimated price of $26.93 for this defensive action. The average respondent's WTP for a reduction in one symptom day from exposure to wildfire smoke is equal to $-26.93/-0.31 = \$86.87$.

Issues such as joint production are often raised when considering the validity of the estimated value for a reduction in symptom days using this method. For instance, if the defensive action used in the estimate enters into an individual's utility function directly, rather than just through its effect on symptom days, WTP values may be inflated. However, a few factors work to our advantage when taking this complication into account. As previously mentioned, respondents were asked to report only those actions taken as a direct result of exposure to the Station Fire smoke. In addition, home air cleaners are typically used for the specific purpose of reducing indoor pollution levels. Therefore, there is good reason to believe that the marginal effect of home air cleaner use on expected symptom days is accurately calculated.

[7] The discrete change in expected count outcome resulting from a change in binary variable X^k from 0 to 1 can be calculated as $[\mu_i | X^k = 0][\exp(\beta^k) - 1]$, where $\mu = \exp(X\beta)$, with all variables except X^k set at their sample mean.

Contingent Valuation Method WTP

One goal of this study is to compare the WTP value estimated by applying the CVM to a simple COI estimate and the WTP value estimated by applying the DBM. Given the assumed log-normal distribution of WTP, the median value can be calculated as

$$E(WTP) = \exp(\mu), \qquad [23]$$

and the expected value as

$$E(WTP) = \exp(\mu + 0.5\sigma^2), \qquad [24]$$

where μ and σ are, respectively, the mean and standard deviation of the logged WTP. Estimates of μ and σ are recovered as follows:

$$E(WTP) = \exp\left[\frac{-\mathbf{x}_i{}'\beta}{\beta_{\text{lnbid}}} + 0.5\left(-\frac{1}{\beta_{\text{lnbid}}}\right)^2\right]. \qquad [25]$$

By setting all independent variables at their sample mean, the WTP to avoid an average number of symptom days experienced in the household can be estimated. This results in a value of $400 to avoid an average of seven symptom days, or $57 per day. Plugging in one symptom day and setting all other independent variables at their average value results in a mean WTP value of $95.03 for a reduction in one wildfire smoke–induced symptom day. This is the value focused on for the comparison across methods. Due to the fact that WTP is increasing at a decreasing rate in symptom days, the WTP to avoid one symptom day is much larger than the WTP per day to avoid an average of seven symptom days. This is consistent with previous studies such as that by Alberini et al. (1997), who found that WTP per day to avoid a five-day illness was about one-third the WTP to avoid a one day illness.

Calculating a median estimate of WTP results in a much smaller value of $8.51 to avoid one day of symptoms. A positively skewed distribution of WTP in this health valuation context has been observed in studies such as those by Loehman et al. (1979) and Dickie et al. (1987). This likely indicates the presence of significant variation in tastes and preferences across the population. We continue to focus on the mean WTP measure of central tendency, given that the majority of previous studies comparing CVM WTP values with either DBM WTP values or COI estimates have focused on mean CVM WTP values to make this comparison (see Rowe and Chestnut 1985; Dickie et al. 1986; Berger et al. 1987; Dickie et al. 1987; Chestnut et al. 1996, 2006), and one focus of this study is comparing our findings to theirs. In addition, efforts to recover total population benefit estimates require sample mean WTP values.

While these estimates represent the WTP for a reduction in 50% of the symptom days that all members of the household experienced, recall that a covariate controlling for the number of people in the household with symptoms was included in the full regression model but was not found to be a significant determinant of the probability that the individual was willing to pay the specified bid amount. Given that having health insurance has a negative and highly significant effect on the probability that the individual is willing to pay the specified bid amount, it would be interesting to compare WTP values for those individuals with and without insurance. However, given the very small sample size of respondents without insurance who were eligible to respond to the stated preference question, we do not make this comparison here.

Cost of Illness

To calculate a traditional COI estimate (COI_{trad}) from the sample of survey respondents, those individuals who experienced health symptoms from exposure to the Station Fire smoke are the focus for the analysis. The direct and indirect costs incurred by each individual as a result of these symptoms are totaled. This includes expenditures on medical visits and prescribed medications, expenditures on nonprescription medications, expenditures on visits to a nontraditional healthcare provider, lost wages, as well as the opportunity cost of time spent obtaining medical care, valued at the individual's wage rate. For each individual, this total COI is divided by the number of symptom days experienced to arrive at a daily COI estimate. Taking the average across the sample of respondents who experienced symptoms, results in a COI per symptom day of $3.02, with a median estimate of $0. This is consistent with the discrepancy between mean and median CVM WTP estimates. It should be noted that the medical costs used in these calculations represent private, out-of-pocket costs of treatment paid by the patient and do not include costs incurred by insurance providers.

Taking this calculation one step further to arrive at a comprehensive COI estimate (COI_{comp}), the value of lost recreation is incorporated into the estimate. For each individual who experienced health symptoms from wildfire smoke exposure, the number of days of recreation that individual missed as a direct result of these symptoms is multiplied by the average consumer surplus per day of sightseeing in the Pacific coast region, estimated to be $20.27 based on four previous study estimates (see Loomis 2005). While there are many estimates of the value of various recreational activities provided in the literature, the activity of sightseeing was chosen for a number of reasons. First, the Public Opinions and Attitudes Survey of 2007 found that in general Californians participate in recreation activities that are inexpensive and require little equipment and technical expertise. Of the top 15 recreational activities participated in, sightseeing was ranked second by participation of those individuals surveyed (California

Department of Parks and Recreation 2009). While walking for fitness or pleasure was ranked first, there are no value estimates for this recreational activity found in the literature. Second, the average age of the Station Fire survey respondents that experienced health effects and lost recreation days as a result was 54 years with a maximum age of 89. Therefore, sightseeing is likely a reasonable activity to capture the lost recreation value to the full range of survey respondents. Incorporating this value into each individual's total COI and dividing by the number of symptom days experienced results in an average comprehensive COI per symptom day of $16.87 (median of $13.51).

TABLE 8
Values for Reduction in One Wildfire Smoke–Induced Symptom Day

Method	Point Estimate	90% CI
COI_{trad}	$3.02	[$1.63–$4.41]
COI_{comp}	$16.87	[$14.11–$19.62]
Defensive behavior WTP	$86.87	[$76.56–$443.26]
Contingent valuation WTP	$95.03	[$22.78–$610.42]

Comparison of Values

The mean CVM WTP value and marginal DBM WTP value are around 30 times larger than the traditional COI estimate and more than five times larger than the comprehensive COI estimate for a reduction in one symptom day. These ratios of WTP:COI are greater than that found in the majority of previous studies that have compared the two. However, this is the only study to calculate this ratio for the specific case of wildfire smoke exposure using primary data. In addition, while only 20 individuals surveyed sought traditional or nontraditional medical care as a result of symptoms, 366 of these individuals took costly defensive actions to protect themselves from exposure to the wildfire smoke. Further, of the 156 respondents who experienced health symptoms, 110 of them missed recreation days as a result of these symptoms. The disutility associated with symptoms or lost recreation captured in the WTP estimate but not the traditional COI estimate may be substantial for individuals exposed to wildfire smoke. Similar to what theory would predict, incorporating the value of lost recreation into the COI estimate reduces the ratio of WTP to COI considerably.

The daily WTP values of $86.87 and $95.03 fall within those estimated in the literature for other air pollutants. Johnson, Fries, and Banzhaf (1997) summarized a number of studies estimating WTP values for a reduction in various health symptoms and found that they ranged from about $5 for a reduction in one day of chest congestion up to about $194 for a reduction in one day of angina symptoms. By combining a meta-analysis of morbidity valuation studies with a health-status index, the authors themselves estimated values from $36 to $68 to avoid one day of mild cough, $110 to avoid one day of shortness of breath, and $91 to $129 to avoid one day of severe asthma.[8]

To explore whether our point estimates of WTP and COI are statistically different, Table 8 presents the estimates of the value for a reduction in one wildfire smoke–induced symptom day, along with the 90% percentile confidence intervals from 1,000 bootstrapped coefficients for the WTP models and standard confidence intervals around the mean COI estimates. The 90% confidence interval of $1.63 to $4.41 around the traditional COI estimate does not overlap the 90% confidence intervals around the WTP values. The null hypothesis that this estimate equals either of the WTP values can be rejected at the 90% confidence level. While the confidence interval around the comprehensive COI estimate of $14.11 to $19.62 is much closer to overlapping the interval around the mean CVM WTP estimate, the null hypothesis that this COI estimate equals either of the WTP values can also be rejected at the 90% confidence level. However, in making this comparison between WTP and COI estimates, a few properties of the WTP measures warrant further discussion. As explained by Haab and McConnell (2002), three sources of randomness in a WTP measure may be present: randomness of preferences, randomness of estimated parameters,

[8] All values were converted to 2009 U.S. dollars using the Consumer Price Index.